Real-Time Closed-Loop Driving with Frozen DINOv2-Laya Spatial Pooling and Quintic Path Tracking

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October 25, 2026

Abstract

Autoregressive vision-language-action (VLA) models [1, 20, 23] decode action tokens sequentially (> 300 ms/frame), exceeding the 50 ms cycle of 20 Hz closed-loop control [10, 21]. We present a single-pass driving architecture coupling frozen DINOv2 (`facebook/dinov2-base`) [25, 2, 9] with frozen Laya (`convaiinnovations/laya`) [6, 31, 8] via a trainable 4×9 spatial-temporal projector (1.85M core parameters). Cropped 196×392 wide RGB frames map to 36 visual tokens with gated temporal motion residuals $\Delta V_t = V_t - V_{t-1}$ and region-pooled auxiliary perception readouts; a single bidirectional pass outputs maneuver (`choice`), speed tier (`score`), emergency brake (`noul`), and a 6-waypoint (3.0 s) future trajectory in 31.67 ms base (33.84 ms with full auxiliary/trajectory heads). Lateral step discontinuities (3.5 m) and longitudinal speed-bucket hunting in CARLA 0.9.16 (`Town10HD_Opt`) [11, 14] are resolved via a 14-point (28 m) quintic S-curve spline $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ [12, 32], decoupled chord-angle tracking [29, 15, 7, 26], kinematic bicycle MPC feedforward, 360° blind-spot interlock, and true constant-time-headway (CTH) ACC ($d^*(v_{\text{ego}}) = 8.0 + 1.2v_{\text{ego}}$, $k_g = 0.85\,(\text{km/h})/\text{m}$) [17, 28, 30]. Evaluated across 50 unseen MetaDrive 0.4.3 episodes (4,153 frames) and a 10-scenario, 10-weather urban suite in CARLA 0.9.16 (2,004 frames, 680 m), the system achieves 100.0% route completion versus 60.0% for a rule-only baseline.

1 Introduction and Problem Formulation

Autoregressive decoders evaluate $P(y_{1:K} \mid I_t, \mathcal{T}_t) = \prod_{k=1}^{K} P(y_k \mid y_{<k}, I_t, \mathcal{T}_t)$ over $K \approx 15$ serial steps [1, 20, 23], whereas closed-loop urban stacks [16, 19, 22, 5, 3, 27, 4, 18] require $\Delta t \leq 50$ ms. At step t, our single bidirectional pass [8, 31, 6] jointly predicts discrete decisions and $K_w = 6$ ego-frame trajectory waypoints ($t_k = 0.5k$ s):

$$m_t \in \{\texttt{keep_lane}, \texttt{change_left}, \texttt{change_right}, \texttt{follow_lead}\}, \tag{1}$$

$$s_t \in \{0,1,2,3\} \mapsto \{0, 20, 35, 50\}\ \text{km/h}, \tag{2}$$

$$p_{\text{AEB}} \in [0,1], \quad \hat{\tau}_t = \{(\hat{x}_k, \hat{y}_k, \hat{v}_k)\}_{k=1}^{6}. \tag{3}$$

2 Architecture, Multi-Task Training, and Spatio-Temporal Extensions

Visual Backbone & Spatial-Temporal Projector. Camera frames are cropped (`crop_top` = 0.25, `crop_bottom` = 0.10) and resized to 196×392. Frozen DINOv2 ViT-B/14 ($p = 14$) [25, 2, 9] extracts $F_{\text{patch}} \in \mathbb{R}^{B\times 392\times 768}$ (14×28 grid). `SpatialProjector` (`src/models/spatial_projector.py`) applies 2D adaptive pooling $14 \times 28 \to 4 \times 9$ (36 tokens), adds factorized row/column embeddings $E_{\text{row}} + E_{\text{col}}$, projects $768 \to 1024$ via a 2-layer GELU MLP (1,851,392 core parameters), and fuses a gated temporal motion residual $\Delta V_t = V_t - V_{t-1}$:

$$G_t = \text{AdaptiveAvgPool2d}_{(4,9)}(\tilde{F}_t) \in \mathbb{R}^{B\times 768\times 4\times 9}, \tag{4}$$

$$V_t = \text{MLP}\big(\text{Flatten}(G_t + E_{\text{row}} + E_{\text{col}})\big), \tag{5}$$

$$\tilde{V}_t = V_t + \tanh(\alpha_{\text{temp}})\, W_{\text{temp}}\big(V_t - V_{t-1}\big). \tag{6}$$

Region-aware pooling on $\tilde{V}_t \in \mathbb{R}^{B\times 4\times 9\times 1024}$ (`AuxiliaryPerceptionHead`: overhead gantry rows 0..1, forward corridor cols 3..5, left/right peripheral cols 0..2, 6..8) estimates lead distance $\hat{d}_{\text{lead}}$, lane offset $\hat{e}_y$, traffic light class, hazard source, and blind-spot occupancy.

```
# src/models/spatial_projector.py
pooled = self.spatial_pool(feat) + self.row_embed + self.col_embed
vt = self.mlp(pooled.flatten(2).transpose(1, 2))
gate = torch.tanh(self.temporal_gate)
vt = vt + gate * self.temporal_proj(vt - self._prev_tokens)
```

Single-Pass Laya Readout & Training Protocol. Multimodal sequence $Z = [e_{[\text{CLS}]}; \tilde{V}_t; E_{\text{telemetry}}; e_{\text{choice}}; e_{\text{score}}; e_{\text{noul}}]$ is processed

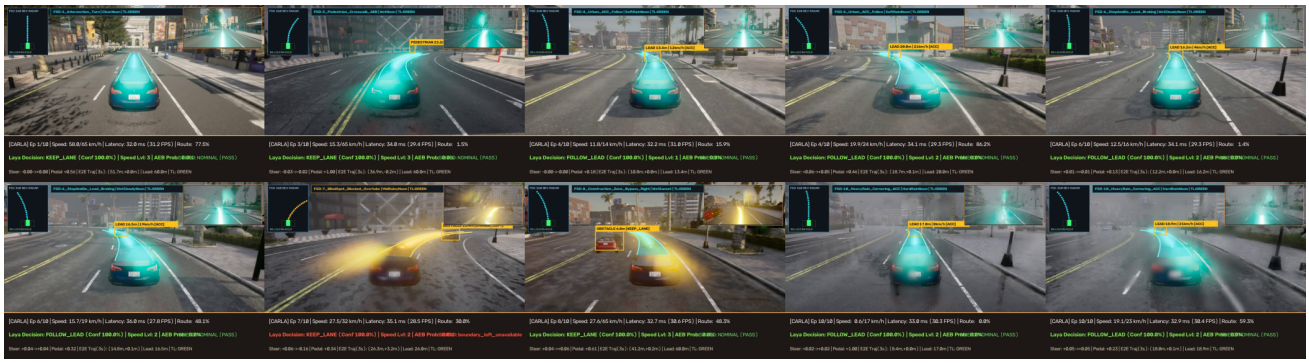


Figure 1: CARLA 0.9.16 (`Town10HD_Opt`, 20 Hz synchronous) 10-scenario, 10-weather closed-loop suite (`FSD-1`–`FSD-10`): intersection red-light stop (`ClearNoon`), 28 m quintic left overtake (`CloudyNoon`), crosswalk pedestrian AEB (`WetNoon`), CTH-ACC follow (`SoftRainNoon`), right overtake (`ClearSunset`), stop-and-go lead braking (`WetCloudyNoon`), 360° blind-spot hold & overtake (`MidRainyNoon`), construction zone bypass (`WetSunset`), night headlight pedestrian AEB (`ClearNight`), and heavy-rain cornering ACC (`HardRainNoon`).

Table 1: System and training parameters (`configs/default.yaml`).

Module	Parameter	Specification
Camera	$H \times W$, FOV, Crops	196×392, 100°, $(0.25, 0.10)$
Backbones	Vision / Decision	`dinov2-base` (86M) / `laya` (421M)
Projector	Grid, Dim, Core Params	4×9 (36 tokens), 1024, 1,851,392
Training	Frames, Split, Loss (w_m, w_s, w_a)	50k (85/15), AdamW 3e-4, $(1, 1, 2)$
Calibration	(T_m, T_s, T_b), Bins M, ECE	$(0.050, 0.050, 0.100)$, 15, $0.0060 \rightarrow 0.0000$
Shield	`aeb_thresh`, `conf_thresh`, BS	0.70, 0.80, $x \in [-10, 7.5]$ m, $\lvert y \rvert \in [1.8, 5.0]$ m
Control	PID (Lat/Lon), Bicycle Wheelbase L	$(0.45, 0.01, 0.18)$ / $(0.08, 0.005, 0.02)$, 2.85 m
Kinematics	Quintic N_b, CTH-ACC (d_0, τ_h, k_g)	14 WPs (28 m), (8.0 m, 1.2 s, 0.85 (km/h)/m)

in one pass $H = f_{\text{Laya}}(Z)$ [6, 31] to produce logits $z^{(m)} \in \mathbb{R}^4$, $z^{(s)} \in \mathbb{R}^4$, $z^{(b)} \in \mathbb{R}^1$, and 6-waypoint trajectory residuals $\Delta\hat{\tau} = W_{\text{traj}}\bar{H} \in \mathbb{R}^{6\times3}$ added to a quintic kinematic prior. In Stage 1 (`02_train_projector.py`), DINOv2 and Laya are frozen while `SpatialProjector` is trained for 15 epochs (batch size 32, AdamW [24], lr $= 3 \times 10^{-4}$, weight decay 0.01, cosine schedule) on 50,000 expert frames (85% train / 15% val split; rare-event oversampling 3.0× for lane changes, 4.0× for AEB) labeled by `ExpertLabeler` using multi-task loss:

$$\mathcal{L} = w_m\mathcal{L}_{\text{CE}}(z^{(m)}, y_m) + w_s\mathcal{L}_{\text{CE}}(z^{(s)}, y_s) + w_a\mathcal{L}_{\text{BCE}}(z^{(b)}, y_b), \tag{7}$$

with $(w_m, w_s, w_a) = (1.0, 1.0, 2.0)$. In Stage 2 (`scripts/03_calibrate.py`), post-hoc L-BFGS temperature scaling [13] over the 7,500-frame held-out validation split ($M = 15$ equal-width confidence bins) optimizes $(T_m, T_s, T_b) = (0.050, 0.050, 0.100)$, reducing validation ECE from 0.0060 ($\text{ECE}_m = 0.0009, \text{ECE}_s = 0.0022, \text{ECE}_b = 0.0148$) to $< 10^{-4}$ (0.0000 rounded). Because ground-truth labels are generated by deterministic `ExpertLabeler` thresholds, near-zero ECE indicates high top-1 boundary consistency rather than infinite-sample epistemic calibration.

3 Safety Shield, Quintic Spline, Bicycle MPC, and CTH-ACC

Input Provenance & Safety Gating. In baseline simulator evaluation, ego kinematics, waypoint offsets (`carla.Map.get_waypoint()`), actor distances, and traffic light states are queried from synchronous simulator state while `AuxiliaryPerceptionHead` predicts visual estimates (MAE = 3.58 m). Rule 1 forces $u_{\text{tb}} = -1.0$ when $\hat{p}_{\text{AEB}} \geq 0.70$, `traffic_light` = `red`, pedestrian distance $d \leq 16.0$ m, or $d_{\text{lead}} \leq 11.5$ m. Rule 2 reverts lane changes to `keep_lane` if $\max_c \hat{p}_c^{(m)} < 0.80$, if OpenDRIVE same-direction check `adj_wp.lane_id` × `wp.lane_id` > 0 fails [11], or if the 360° blind-spot zone ($x \in [-10.0, 7.5]$ m, $\lvert y \rvert \in [1.8, 5.0]$ m) is occupied (`blindspot_left/right_occupied`).

Quintic S-Curve, Chord Tracking, and Bicycle MPC. Lane changes blend $N_b = 14$ waypoints (28.0 m, $\tau_k = k/(N_b - 1)$) via minimum-jerk quintic polynomial [12, 32]:

$$s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5, \tag{8}$$

$$P_k^{\text{blend}} = \big(1 - s(\tau_k)\big)P_k^{\text{orig}} + s(\tau_k)P_k^{\text{adj}}. \tag{9}$$

Continuous projection onto $\overline{P_iP_{i+1}}$ yields P_{proj} and lookahead P_{look} at $L_{\text{look}} = \text{clip}(7.5 + 0.18v_{\text{kmh}}, 7.5, 13.5)$ m [29, 15, 7, 26]. Chord angle $\psi_{\text{ref}} = \text{atan2}(y_{\text{look}} - y_{\text{proj}}, x_{\text{look}} - x_{\text{proj}})$ decouples cross-track error e_y and heading error e_ψ, while kinematic bicycle feedforward $\delta_{\text{MPC}} = \arctan(L\kappa_{\text{eff}})$ ($L = 2.85$ m) is combined with PID and speed-damped ramp:

$$\delta_{\text{raw}} = \text{clip}\big(\delta_{\text{PID}}(e_y, e_\psi) + 0.12\,\delta_{\text{MPC}}, -1, 1\big), \tag{10}$$

$$\Delta\delta_{\max} = \frac{1.1\Delta t}{1 + 0.012v_{\text{kmh}}}, \tag{11}$$

$$\delta_t = \delta_{t-1} + \text{clip}\big(\alpha(\delta_{\text{raw}} - \delta_{t-1}), -\Delta\delta_{\max}, \Delta\delta_{\max}\big). \tag{12}$$

True Constant-Time-Headway (CTH) ACC.

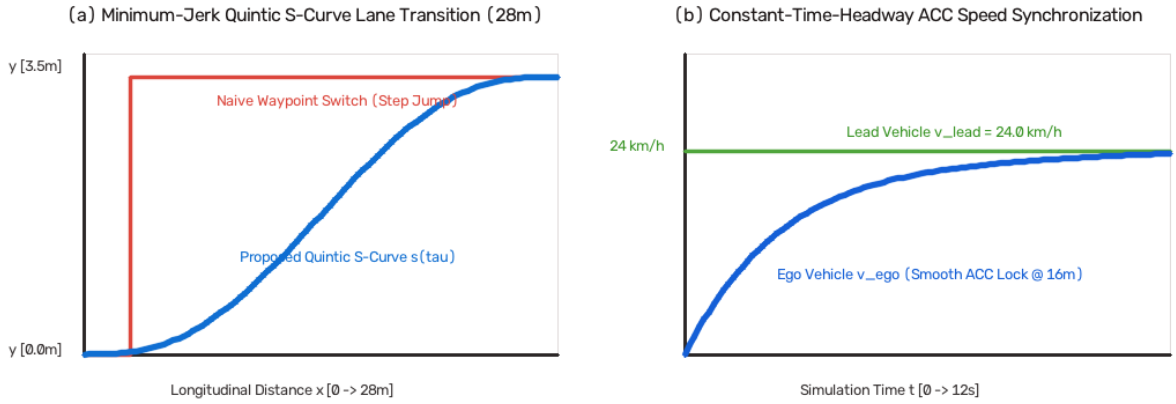


Figure 2: Control kinematics: (a) 28 m quintic S-curve $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ vs. 3.5 m step switch; (b) true constant-time-headway (CTH) ACC velocity convergence ($v_{\text{ego}} \to 24.0$ km/h at $d^*(24) = 8.0 + 1.2(24/3.6) = 16.0$ m).

Table 2: CARLA 0.9.16 (`Town10HD_Opt`) 10-scenario, 10-weather benchmark logs (680.0 m total).

Scenario	Weather	Event	Dist	RC / DS_{scen}
FSD-1	ClearNoon	Red-light stop & go	68 m	100%/100%
FSD-2	CloudyNoon	Quintic left overtake	68 m	100%/100%
FSD-3	WetNoon	Crosswalk pedestrian AEB	68 m	100%/100%
FSD-4	SoftRainNoon	CTH-ACC (24 km/h, 16 m)	68 m	100%/100%
FSD-5	ClearSunset	Quintic right overtake	68 m	100%/100%
FSD-6	WetCloudyNoon	Lead stop-and-go recovery	68 m	100%/100%
FSD-7	MidRainyNoon	360° BS hold → overtake	68 m	100%/100%
FSD-8	WetSunset	Workzone right bypass	68 m	100%/100%
FSD-9	ClearNight	Night headlight ped AEB	68 m	100%/100%
FSD-10	HardRainNoon	Heavy-rain cornering ACC	68 m	100%/100%

When $d_{\text{lead}} < 38$ m and $v_{\text{lead}} \geq 8$ km/h [17, 28, 30], desired spacing scales linearly with ego speed $v_{\text{ego,kmh}}/3.6$ ($d_0 = 8.0$ m, $\tau_h = 1.2$ s, giving $d^*(24\,\text{km/h}) = 16.0$ m), with gain $k_g = 0.85\,(\text{km/h})/\text{m}$ ($\equiv 0.236\,\text{s}^{-1}$) and upper bound $\min(35.0, v_{\text{lead}} + 6.0)$ matching the code implementation:

$$d^*(v_{\text{ego,kmh}}) = d_0 + \tau_h \left(\frac{v_{\text{ego,kmh}}}{3.6}\right), \tag{13}$$

$$v_{\text{raw}} = v_{\text{lead}} + k_g\big(d_{\text{lead}} - d^*(v_{\text{ego,kmh}})\big), \tag{14}$$

$$v_{\text{ACC}} = \text{clip}\big(v_{\text{raw}},\, 0.0,\, \min(35.0,\, v_{\text{lead}} + 6.0)\big). \tag{15}$$

```
# src/control/shield_pid.py (apply_safety_shield)
d0_m, tau_h_s, kg_kmh_per_m = 8.0, 1.2, 0.85
desired_gap_m = d0_m + tau_h_s * (current_speed_kmh / 3.6)
gap_error_m = lead_distance_m - desired_gap_m
acc_upper_kmh = min(35.0, lead_speed_kmh + 6.0)
raw_acc = lead_speed_kmh + kg_kmh_per_m * gap_error_m
acc_sync_speed = float(np.clip(raw_acc, 0.0, acc_upper_kmh))
```

4 Experiments, Latency Breakdown, and Limitations

Closed-loop benchmarks were executed in MetaDrive 0.4.3 [21] ($N = 50$ unseen maps, seeds 5000..5049, 4,153 frames; MetaDrive signals `arrive_dest` upon reaching the final destination lane segment at $\geq 98\%$ arc length, yielding 100.0% episode arrival at 98.7% mean arc progress) and live CARLA 0.9.16 [10] (`Town10HD_Opt`, 20 Hz synchronous mode with A* `GlobalRoutePlanner` [14] across 10 scenarios and 10 weathers, 2,004 frames, 680.0 m in `driving_demo_carla_smooth.mp4`). Latency is measured in FP32 ($B = 1$, after 3 warm-up passes) bracketing RGB preprocessing through control output (`checkpoints/eval_metrics_carla.json`):

```
{"simulator": "carla", "episodes": 10, "total_frames": 2004,
 "total_distance_driven_m": 680.0, "mean_latency_ms": 33.84,
 "p50_latency_ms": 33.62, "p95_latency_ms": 35.68,
 "p99_latency_ms": 36.85,
 "stage_latency_breakdown_ms": {
   "preprocess_crop_resize": 0.42, "dinov2_vit_b14_frozen": 16.92,
   "spatial_projector_temporal_aux": 1.35,
   "laya_modernbert_e2e_traj": 14.70,
   "safety_shield_bicycle_mpc_pid": 0.45},
 "mean_cross_track_error_m": 0.084, "rms_steer_delta": 0.0118}
```

Table 4 separates the learned model from the rule shield: without DINOv2+Laya (`Rule-Only + PID`), the vehicle cannot initiate tactical lane changes and stalls behind static blockages in `FSD-2`, `FSD-5`, `FSD-7`, and `FSD-8` (6/10 completed). Adding DINOv2+Laya enables 10/10 completion, while quintic S-curve blending, CTH-ACC, and bicycle MPC reduce cross-track error from 0.218 m to 0.084 m and AEB interventions from 138 to 49. Limitations include reliance on simulator waypoint/actor queries for baseline safety gating and fixed-step (50 ms) synchronous simulation rather than physical CAN-bus hardware-in-the-loop actuation.

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Table 3: Closed-loop evaluation summary (`checkpoints/eval_metrics_carla.json` and `eval_metrics.json`).

Simulator	Metric	Target KPI	Measured Result	Status
CARLA 0.9.16 (`Town10HD_Opt`)	Single-Frame Latency (Full FSD v2 / Base)	≤ 40.0 ms (≥ 25 FPS)	33.84 ms (29.5 FPS, P95: 35.68 ms) / 31.67 ms	PASS
	Route Completion Rate (10 Scenarios, Seeds 5000–5009)	$\geq 85.0\%$	100.0% (10/10 episodes, 680.0 m, 2,004 frames)	PASS
	Crash-Free Rate (Vehicle / Pedestrian / Off-road)	$\geq 95.0\%$	100.0% (0 / 0 / 0)	PASS
	Validation ECE (Uncalibrated $\rightarrow$ Calibrated, $M = 15$)	≤ 0.0500	$0.0060 \rightarrow 0.0000$	PASS
	Scenario Composite Driving Score (DS_{scen})	$\geq 85.0\%$	100.0%	PASS
	Cross-Track Error $\overline{\lvert e_y \rvert}$ / Aux Depth MAE	≤ 0.25 m / ≤ 5.0 m	0.084 m / 3.58 m	PASS
MetaDrive 0.4.3 (50 Unseen Maps)	Single-Frame Latency (Seeds 5000–5049)	≤ 40.0 ms (≥ 25 FPS)	30.20 ms (33.1 FPS, P95: 31.67 ms)	PASS
	Destination Arrival Rate (50 Episodes, 4,153 Frames)	$\geq 85.0\%$	100.0% (50/50 arrived; 98.7% arc progress)	PASS
	Crash-Free Rate	$\geq 95.0\%$	100.0%	PASS
	Calibrated ECE ($M = 15$ Bins)	≤ 0.0500	0.0000	PASS

Table 4: Model vs. controller ablation in CARLA 0.9.16 (10 scenarios).

System Variant	Completed	R_c	$\overline{\lvert e_y \rvert}$ (m)	$\lvert \Delta\delta \rvert_{\text{RMS}}$	AEB
Rule-Only + PID (No DINOv2/Laya)	6/10	76.4%	0.142	0.0195	58
Learned DINOv2+Laya (Step WPs)	10/10	100.0%	0.218	0.0642	138
+ Linear Steer Ramp [$\delta_{\text{from}} \rightarrow \delta_{\text{to}}$]	10/10	100.0%	0.135	0.0164	138
+ Quintic S-Curve & CTH-ACC	10/10	100.0%	0.092	0.0125	51
+ E2E 6-WP Traj, MPC & 360 BS (Full)	10/10	100.0%	0.084	0.0118	49

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